

Syndicate Exercise

L'or Noir Control System Design Competition

SOLUTION



SOLUTION

SKG16.4 Advanced Process Control/
SKG16.1 Operations Management
Control Theory and Applications Course

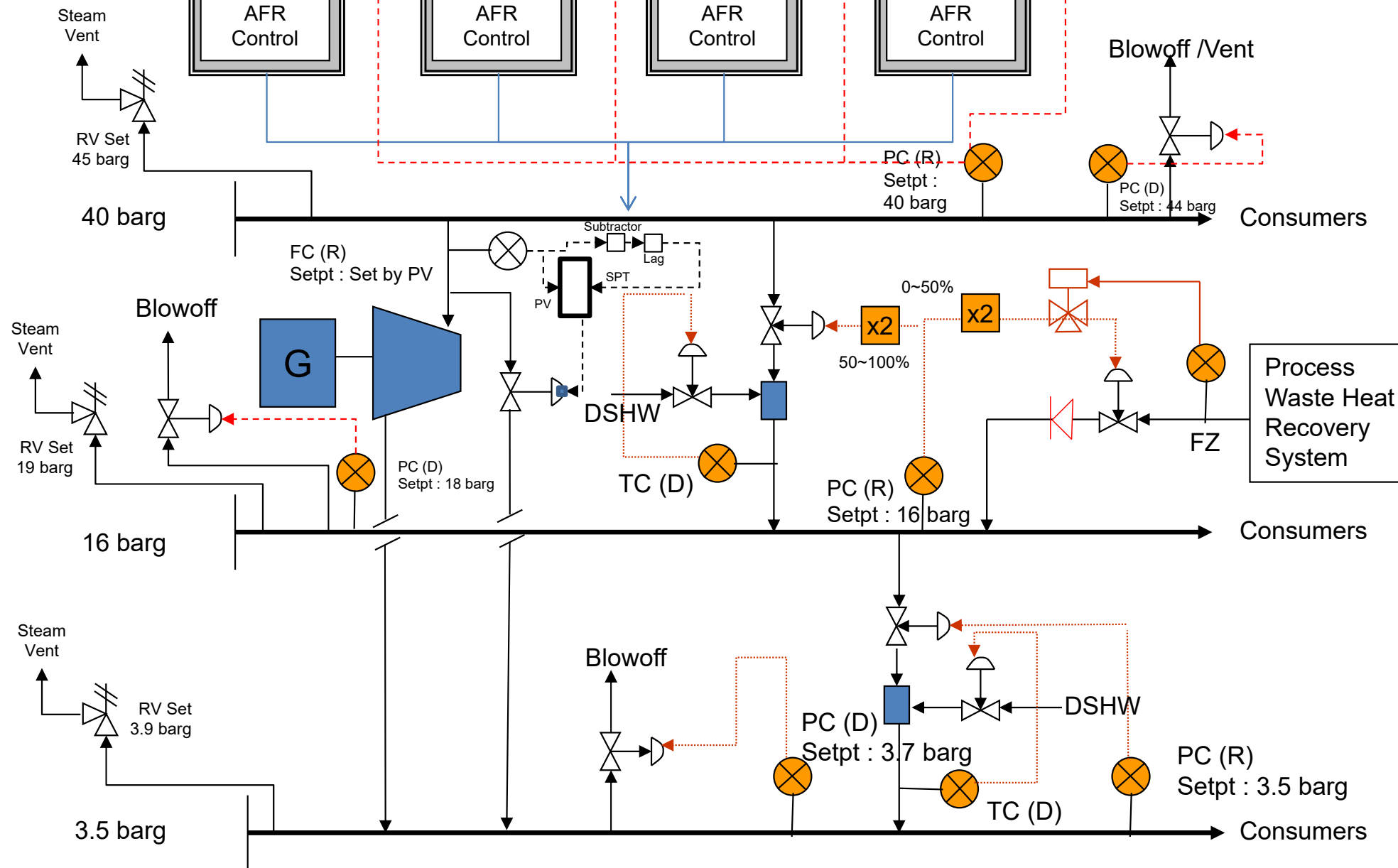
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Revision History



Rev	Date	Prepared By	Notes
2	8 th June	Prakash K.K.	Included letdown stations
3	10 th October 05	Prakash K.K.	Correction for LP blow off controls
4	17 Mar 2011	Andriny R.	Addition to exhaust of turbine
5	21 Oct 2011	Andriny R.	Included stream description
6	22 nd Aug 2012	Prakash K.K.	Updated for SKG16.1
7	22 nd Aug 2012	Prakash K.K.	Start-up scenario amended
7A	22 nd Aug 2012	Prakash K.K.	Amended for SKG16.1
9	12 th Feb 2014	Prakash K.K.	Minor Amendments
47	10 th April 2017	Prakash K.K.	Alignment of Revision to Main Module and enhancement to answers to include STG bypass
47a	21 st April 2017	Prakash K.K.	Addition of Basic Engineering - Overview of Solution slide
47g	07 th Sept 2018	Prakash K.K.	Relocation of Check Valve and addition of blow-offs



Basic Engineering – Solution Overview



- The basic design/approach for this solution shall be as follows:
 - Each header represents a **distribution control** i.e. supply/demand control
 - Supply/Demand control (as indicated in the Learning Module) by a Pressure transmitter and control on each header
 - The final handle for each of the header-controls, shall be as follows:
 - HP header : final handle is the boilers via the individual AFR
 - MP header : two handles, the HP/MP letdown and WHRU steam
 - LP header : one handle : MP/LP letdown (note, that STG flow is controlled by electricity demand and cannot be controlled as per project specification)
 - Steam entering each header shall be at the **correct superheat** and temperature sensors and controls will be located immediately downstream of the desuperheater
 - In the event of the **turbine trip**,
 - two key scenarios that need to be managed:
 - High pressure in the HP header (due to excess steam)
 - Low pressure in the LP header (due to loss of steam from STG)
 - Two options are present
 - Monitoring the pressure in the HP header and use a high pressure controller and if the PV goes too high, to open the letdown valve until the pressure normalizes and then close it back
 - Using a flow controller on the common inlet to the STG and if this drops too low, to open the valve and slowly close it back
 - In general, dynamics of flow is always faster than pressure. As such, flow control is used.

Control Description ...1



- **HP header**

- The HP header pressure can be controlled by a pressure controller located on the HP header and connected to the individual boiler A/F ratio control system
- Increases in demand in the steam system (either HP/MP/LP) will eventually impact steam draw from the HP system and this will be replenished by the boilers via the PC control loop.

- **MP header**

- The MP header will primarily be controlled by a pressure control on the MP header which controls the steam from the Waste Heat Recovery system.
- At max process production (HOLD), If steam is sufficient, then the controller will throttle the waste heat steam flow. Temperature is also controlled at that side
- At max process production (HOLD), if steam is insufficient, the control system will open (via a split range) the HP/MP letdown
- The HP/MP letdown has a desuperheater station that has a temperature controller that will manipulate the amount of water (BFW) in order to control the superheat temperature of the MP steam

Description ...2



- ***MP header (contd)***
 - Steam can also be prevented from backflow of MP steam into the Waste Heat unit by the installation of a check-valve
 - The check valve shall be backed up by an Instrumented Protective Function (Low Flow trip) which shall trip the waste steam feed valve upon low flow condition (this decision to install both is based on both HAZOP and IPF classification requirements).
- **LP header**
 - The LP header will be controlled by a pressure controller set at 3.5 barg which controls the MP/LP letdown which will in turn supplement the steam received from the backpressure turbine
 - The LP header is allowed to increase in pressure till 3.7 barg at the setting of the second high pressure controller on the LP header.
 - Similar to the HP/MP letdown, a temperature control controls the superheat temperature ex the MP/LP letdown station by manipulating BFW to the desuperheater



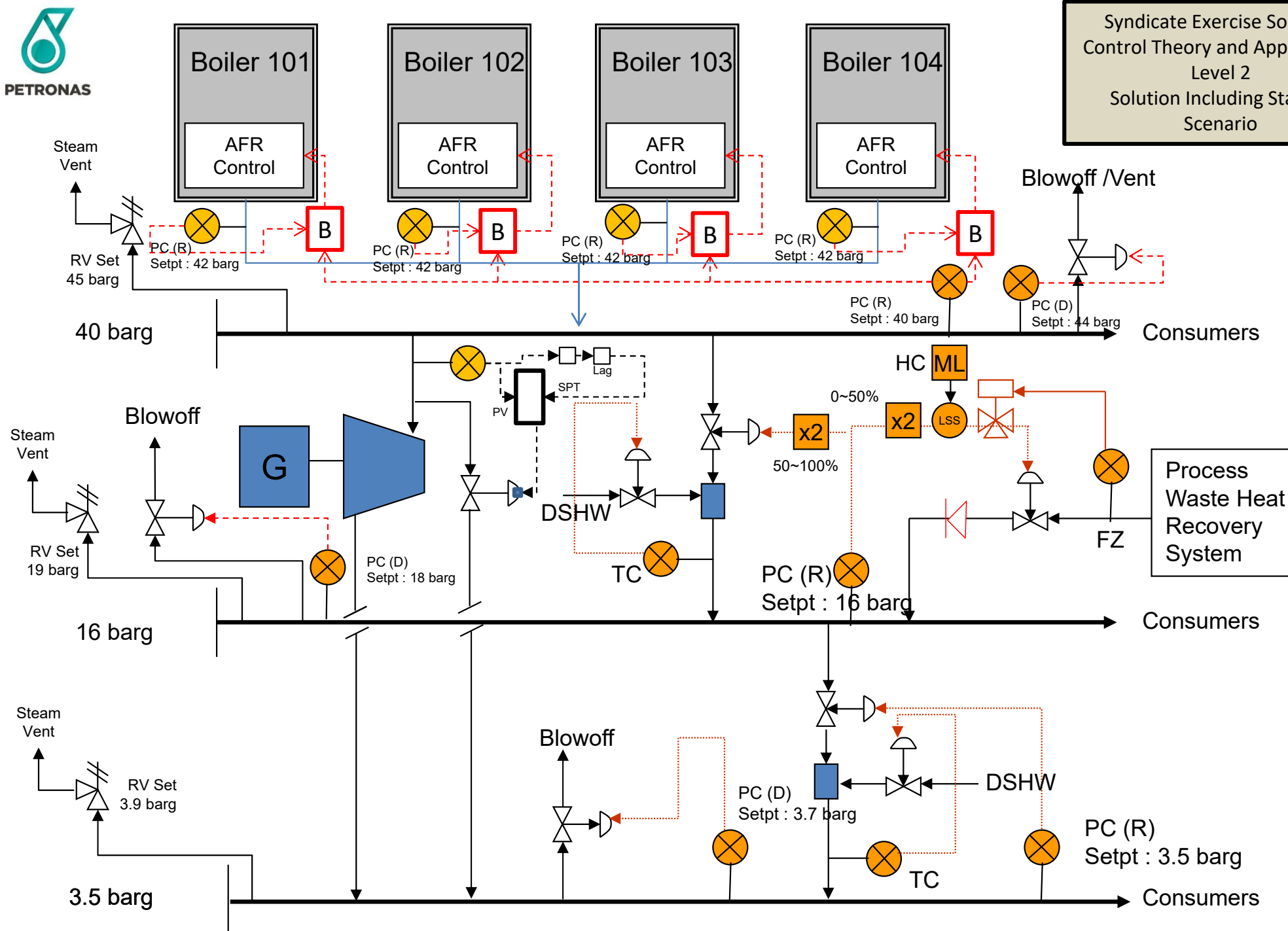
Startup scenario

- It should be noted that one of the first units to start-up in the plant (typically after power, instrument air and fuel gas) is the steam system. This is to provide steam for process drivers (steam turbines) and process heaters.
- It should also be noted that typically waste heat recovery units are found in the process units. These are therefore started much later than the main steam systems.
- HP System
 - Boilers are started individually and the HP steam pressure control system shall be configured to be placed on line with any number of boilers (typical requirement)
 - As new boilers come on line, this boiler's combustion control scheme shall be allowed to connect to the master header pressure control system
 - Bias stations are provided for each boiler to allow this feature
- MP System
 - In order to cover start-up scenarios where the waste heat unit cannot send steam (because the process unit is still down), the valve from the waste heat unit will have to be shut.
 - With the MP header controller in Automatic, the MP controller will be requesting for maximum opening of this valve. It will not be possible to line up the waste steam system with this valve fully open.
 - To cater for this situation, a manual loader station (or handstation) can be utilised together with a low signal selector to ensure that the valve is closed during start-up.

Trip of GT scenario



- If the power turbine trips, there will be a severe reduction of LP steam supply.
- The LP letdown will immediately open to tackle this and will cause an abrupt pressure drop of LP header causing the MP-LP letdown to open.
- The function of the controller at the GT is to minimise the upset
 - It will momentarily supply to LP header while allowing the 2 MP and LP header pressure to stabilise



Other alternatives



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- Implementation of protective controllers when header pressures hit high or low limits by closing/opening the letdown stations
 - TC on GTG letdown when it trips



Last Slide

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